



Department of Electrical and Electronics Engineering Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. EEE

Course Code & Title: EE3020 Smart System Automation

Name of the Faculty member: Dr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic: Unit I / Smart sensors - Types**
- **Course Outcome: CO 1**
- **Topic Learning Outcome: 1c**
- **Activity Chosen: Mini map**
- **Justification:**

Mini mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mini mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches. For this topic, there are different types of smart sensors and its characteristics can be brought in to single map for better understanding.

- **Time Allotted for the Activity: 10 minutes**
- **Details of the Implementation:**

Mini mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mini map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

- In my subject Mini map was conducted for the topic smart sensors. The students are voluntarily drawn the map.

- **CO – PO / PSO mapping:**

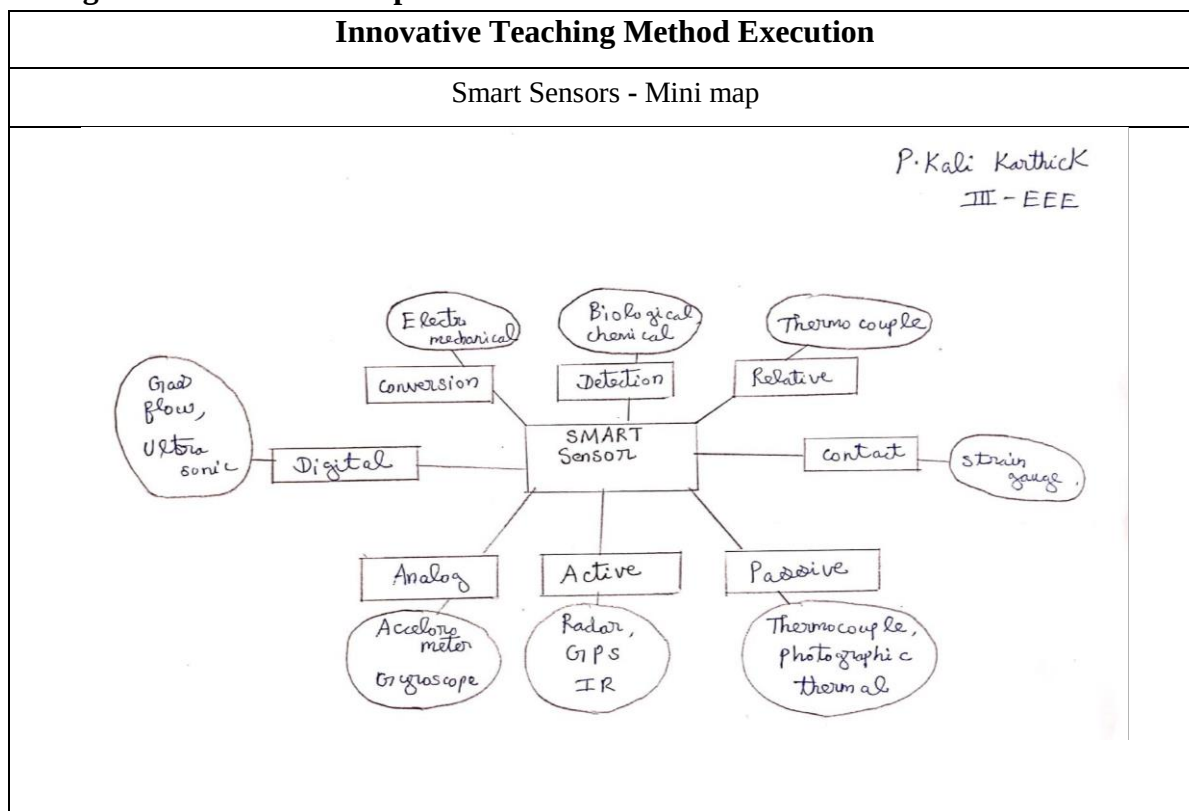
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	-	-	-	1	2	1	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The benefits of Mini Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mini map made slow learners to remember the information more quickly.

Challenges faced in implementation:

Normally teachers will give longer explanations in the notes section of the topic. The students are made into groups and to draw the map to indicate relationships between the topics in biomass conversion process. I planned the activity for 10 minutes only. But in real scenario it takes 15 minutes to complete this activity.

References:

1. Grimm, Christoph, Neumann, Peter, Mahlkech and Stefan, Embedded Systems for Smart Appliances and Energy Management, Springer 2013, 1st Edition.
2. KazemSohraby, Daniel Minoli and TaiebZnati, Wireless Sensor Networks Technology, Protocols, and Applications, John Wiley & Sons, 2007, 1st Edition.
3. NilanjanDey, Amartya Mukherjee, Embedded Systems and Robotics with Open-Source Tools, CRC press, 2016, 1st Edition.

Signature of Faculty Member

HOD



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Course Code & Title: EE3020 Smart System Automation

Name of the Faculty member (s): Dr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit II / Sensing Requirements
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** 2c
- **Activity Chosen:** One minute paper
- **Justification:**

Using a one-minute paper for the topic of sensing requirements is effective because it allows students to quickly reflect on and consolidate their understanding, promoting active learning and critical thinking. It helps identify knowledge gaps, enabling instructors to provide immediate clarification. This method offers efficient feedback, ensuring both students and instructors can address misconceptions promptly. Overall, it enhances engagement and reinforces key concepts effectively.

Time Allotted for the Activity: 5 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content, which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 2	3	2	1	2	2	-	-	-	1	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution	
Sensing requirement – One minute paper	
THIRUCHARAI G. SENSOR REQUIREMENT Type of measurement: Determine what physical quantity or property you need to measure (eg pressure, humidity, motion) Accuracy: Define the level accuracy required for your measurements. This could be in terms of percentage, absolute value or resolution. Precision: Specify the precision needed for your measurements. Precision refers to the consistency or reproducibility of measurements. Range: Determine the range over which the sensor needs to operate effectively. Sensors have limits to their measurement capabilities so choose one that fits within your required range. Response time: Consider how quickly the sensor needs to respond to change in the measured quantity.	

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept, which was reflected from their answers for the questions I have asked during discussion session.

- ❖ **Benefit of the practice:**

- ✓ Students can able to attend the question even in the questions are in indirect form.
 - ✓ Students can able to explain the concepts in examination without any confusion.

- ❖ **Challenges faced in implementation:**

- ✓ Time utilization for conducting activity.

References:

1. Grimm, Christoph, Neumann, Peter, Mahlkech and Stefan, Embedded Systems for Smart Appliances and Energy Management, Springer 2013, 1st Edition.
2. KazemSohraby, Daniel Minoli and TaiebZnati, Wireless Sensor Networks Technology, Protocols, and Applications, John Wiley & Sons, 2007, 1st Edition.
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Innovative Practice Description

- **Unit / Topic:** Unit III / Energy Measurement Technique
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** 3c
- **Activity Chosen:** Think pair share
- **Justification:**
 - ✓ It helps students to think individually about a topic or answer to a question.
 - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
 - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Think-Pair-Share innovative practice conducted after explained the concept of Energy Measurement Technique. First, I asked each student to think about the role, advantages and disadvantages of energy measurement technique for 2 minutes. Then I make them as a pair up to discuss their thoughts and answers. Encourage them to compare their ideas and clarify any misunderstandings. 3 minutes. Finally I invited the pairs to share their insights with the larger group or the whole class for 10 minutes.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	-	-	-	1	2	1	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution	
Energy Measurement Technique – Think Pair Share	
P. Balasaraswathy A. Lissaha Vardhini <u>Energy Management Techniques:-</u> Energy Management is the term for the process of conserving energy. Every company requires energy management in a every ways. Indeed, the global requirement to conserve energy and reduce operating reduce has ramifications in various other fields <u>Techniques:-</u> 1. Actively Regulate Energy use in real time 2. Take control 3. Take control of your Energy usage <u>Advantages:-</u> 1. Cost Reduction 2. Lowered Carbon Footprint 3. Enhanced Wellbeing & Productivity <u>Disadvantages:-</u> 1. Poor Control performance 2. Difficult to achieve the global optimal solution	

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- ❖ **Benefit of the practice:**

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

- ❖ **Challenges faced in implementation:**

I planned the activity for 10 minutes. But in Class room it takes 15 minutes.

References:

1. Grimm, Christoph, Neumann, Peter, Mahlknecht and Stefan, Embedded Systems for Smart Appliances and Energy Management, Springer 2013, 1st Edition.
2. KazemSohraby, Daniel Minoli and TaiebZnati, Wireless Sensor Networks Technology, Protocols, and Applications, John Wiley & Sons, 2007, 1st Edition.
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Name of the Faculty member: Dr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit IV / Activity Monitoring

- **Course Outcome:** CO4

- **Topic Learning Outcome:** 4e

- **Activity Chosen:** Flipped Classroom

- **Justification:**

This topic is chosen, since it can be easily understood by students. Students can improve their understanding on this topic by this activity. More time for discussion is given the class unlike traditional learning. This activity greatly helps the student to know the concept much deeper.

- **Time Allotted for the Activity:** 25 Minutes

- **Details of the Implementation:**

Each team was asked to do literature survey about the given topic. They should include at least 5 Reference material about the given topic for which the material is already shared. Students in a team were asked to present their topics. Queries will be asked by the students to the presenting team

- **CO – PO / PSO mapping:**

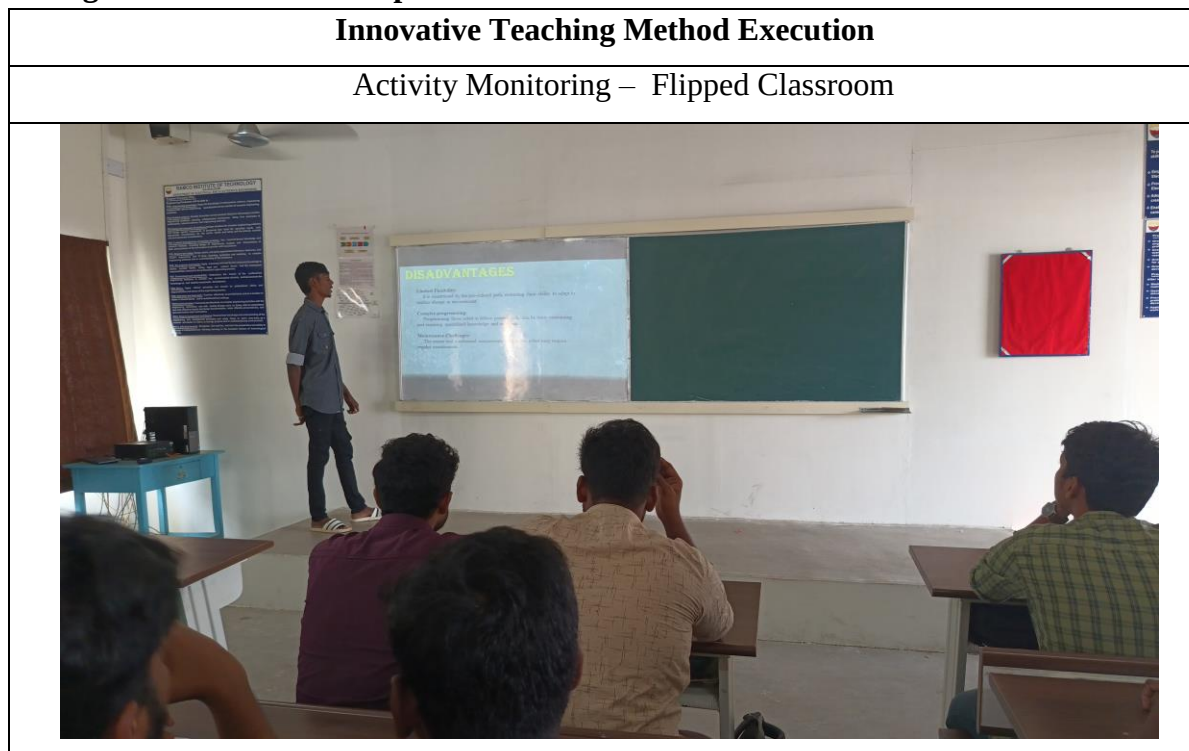
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(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students enjoyed this Flipped classroom Activity. After the class many students requested me to insist other Course faculty members to conduct this activity. I have taken this feedback to by Head of the Department

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students actively participated in the flipped classroom activity. As a teacher I could know the real students feeling, their understanding level from this activity. This greatly helped me to clear all the student doubt and make them more involved in the learning process

- ❖ **Challenges faced in implementation:**

Still few students were hesitant to convey their points and queries. Few students have not taken this activity seriously and remain quite in the class

References:

1. Grimm, Christoph, Neumann, Peter, Mahlknecht and Stefan, Embedded Systems for Smart Appliances and Energy Management, Springer 2013, 1st Edition.
2. KazemSohraby, Daniel Minoli and TaiebZnati, Wireless Sensor Networks Technology, Protocols, and Applications, John Wiley & Sons, 2007, 1st Edition.
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Name of the Faculty member: Dr. A. Arun Kumar

Innovative Practice Description

- **Unit / Topic:** Unit V / Embedded processor based: pick and place robot

- **Course Outcome:** CO 5

- **Unit Outcome:** 5d

- **Activity Chosen:** Demonstration

- **Justification:**

The Embedded processor based: pick and place robot is taught using the active learning approach of demonstration, Students' involvement in the learning is improved based on this demonstration approach.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about working of pick and place robot for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the construction of the machine for another 1 minute.
- ✓ Finally, I shown the motor parts for each student and explained.

- **CO – PO / PSO mapping:**

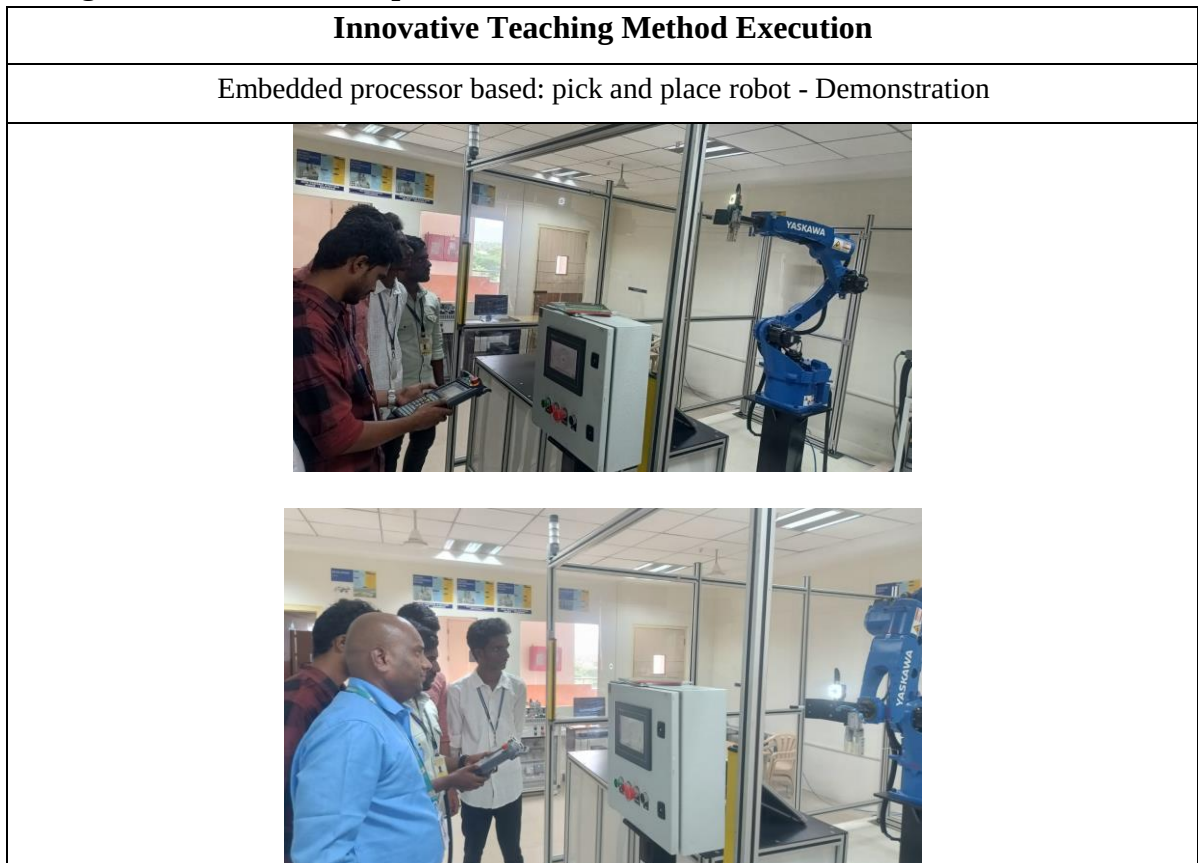
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	-	-	-	1	2	1	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9
	1
Justification for correlation	The students can Function effectively as a team

• **Images / Screenshot of the practice:**



Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students told that it is good see the working of pick and place robot individually and demonstration helps the students to understand the concepts easily.

❖ ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Students can able to explain the concepts in examination without any confusion.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

1. Grimm, Christoph, Neumann, Peter, Mahlkech and Stefan, Embedded Systems for Smart Appliances and Energy Management, Springer 2013, 1st Edition.
2. KazemSohraby, Daniel Minoli and TaiebZnati, Wireless Sensor Networks Technology, Protocols, and Applications, John Wiley & Sons, 2007, 1st Edition.
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